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CSES Problem Set**Josephus Queries**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a game where there are n children (numbered $1, 2, \dots, n$) in a circle. During the game, every second child is removed from the circle, until there are no children left.

Your task is to process q queries of the form: "when there are n children, who is the k th child that will be removed?"

Input

The first input line has an integer q : the number of queries.

After this, there are q lines that describe the queries. Each line has two integers n and k : the number of children and the position of the child.

Output

Print q integers: the answer for each query.

Constraints

- $1 \leq q \leq 10^5$
- $1 \leq k \leq n \leq 10^9$

Example

Input:

```
4
7 1
7 3
2 2
1337 1313
```

Output:

```
2
6
1
1107
```

CSES Problem Set

Exponentiation

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to efficiently calculate values a^b modulo $10^9 + 7$.

Note that in this task we assume that $0^0 = 1$.

Input

The first input line contains an integer n : the number of calculations.

After this, there are n lines, each containing two integers a and b .

Output

Print each value a^b modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq a, b \leq 10^9$

Example

Input:

```
3
3 4
2 8
123 123
```

Output:

```
81
256
921450052
```

CSES Problem Set

Exponentiation II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to efficiently calculate values a^{b^c} modulo $10^9 + 7$.

Note that in this task we assume that $0^0 = 1$.

Input

The first input line has an integer n : the number of calculations.

After this, there are n lines, each containing three integers a , b and c .

Output

Print each value a^{b^c} modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^5$
- $0 \leq a, b, c \leq 10^9$

Example

Input:

```
3
3 7 1
15 2 2
3 4 5
```

Output:

```
2187
50625
763327764
```

CSES Problem Set**Counting Divisors**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given n integers, your task is to report for each integer the number of its divisors.

For example, if $x = 18$, the correct answer is 6 because its divisors are 1, 2, 3, 6, 9, 18.

Input

The first input line has an integer n : the number of integers.

After this, there are n lines, each containing an integer x .

Output

For each integer, print the number of its divisors.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq x \leq 10^6$

Example

Input:

3
16
17
18

Output:

5
2
6

CSES Problem Set**Common Divisors**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n positive integers. Your task is to find two integers such that their greatest common divisor is as large as possible.

Input

The first input line has an integer n : the size of the array.

The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print the maximum greatest common divisor.

Constraints

- $2 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^6$

Example

Input:

5
3 14 15 7 9

Output:

7

CSES Problem Set

Sum of Divisors

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Let $\sigma(n)$ denote the sum of divisors of an integer n . For example, $\sigma(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28$.

Your task is to calculate the sum $\sum_{i=1}^n \sigma(i)$ modulo $10^9 + 7$.

Input

The only input line has an integer n .

Output

Print $\sum_{i=1}^n \sigma(i)$ modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^{12}$

Example

Input:

5

Output:

21

CSES Problem Set

Divisor Analysis

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an integer, your task is to find the number, sum and product of its divisors. As an example, let us consider the number 12:

- the number of divisors is 6 (they are 1, 2, 3, 4, 6, 12)
- the sum of divisors is $1 + 2 + 3 + 4 + 6 + 12 = 28$
- the product of divisors is $1 \cdot 2 \cdot 3 \cdot 4 \cdot 6 \cdot 12 = 1728$

Since the input number may be large, it is given as a prime factorization.

Input

The first line has an integer n : the number of parts in the prime factorization.

After this, there are n lines that describe the factorization. Each line has two numbers x and k where x is a prime and k is its power.

Output

Print three integers modulo $10^9 + 7$: the number, sum and product of the divisors.

Constraints

- $1 \leq n \leq 10^5$
- $2 \leq x \leq 10^6$
- each x is a distinct prime
- $1 \leq k \leq 10^9$

Example

Input:

```
2
2 2
3 1
```

Output:

```
6 28 1728
```

CSES Problem Set

Prime Multiples

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given k distinct prime numbers $a_1, a_2, \dots, a_k$ and an integer n .

Your task is to calculate how many of the first n positive integers are divisible by at least one of the given prime numbers.

Input

The first input line has two integers n and k .

The second line has k prime numbers $a_1, a_2, \dots, a_k$.

Output

Print one integer: the number integers within the interval $1, 2, \dots, n$ that are divisible by at least one of the prime numbers.

Constraints

- $1 \leq n \leq 10^{18}$
- $1 \leq k \leq 20$
- $2 \leq a_i \leq n$

Example

Input:

20 2

2 5

Output:

12

Explanation: the 12 numbers are 2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18, 20.

CSES Problem Set**Counting Coprime Pairs**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a list of n positive integers, your task is to count the number of pairs of integers that are coprime (i.e., their greatest common divisor is one).

Input

The first input line has an integer n : the number of elements.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the list.

Output

Print one integer: the answer for the task.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq x_i \leq 10^6$

Example

Input:

8
5 4 20 1 16 17 5 15

Output:

19

CSES Problem Set**Next Prime**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a positive integer n , find the next prime number after it.

Input

The first line has an integer t : the number of tests.

After that, each line has a positive integer n .

Output

For each test, print the next prime after n .

Constraints

- $1 \leq t \leq 20$
- $1 \leq n \leq 10^{12}$

Example

Input:

```
5
1
2
3
42
1337
```

Output:

```
2
3
5
43
1361
```

CSES Problem Set

Binomial Coefficients

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to calculate n binomial coefficients modulo $10^9 + 7$.

A binomial coefficient $\binom{a}{b}$ can be calculated using the formula $\frac{a!}{b!(a-b)!}$. We assume that a and b are integers and $0 \leq b \leq a$.

Input

The first input line contains an integer n : the number of calculations.

After this, there are n lines, each of which contains two integers a and b .

Output

Print each binomial coefficient modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^5$
- $0 \leq b \leq a \leq 10^6$

Example

Input:

```
3
5 3
8 1
9 5
```

Output:

```
10
8
126
```

CSES Problem Set**Creating Strings II**TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

Given a string, your task is to calculate the number of different strings that can be created using its characters.

Input

The only input line has a string of length n . Each character is between a–z.

Output

Print the number of different strings modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^6$

Example

Input:
aabac

Output:
20

CSES Problem Set**Distributing Apples**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n children and m apples that will be distributed to them. Your task is to count the number of ways this can be done.

For example, if $n = 3$ and $m = 2$, there are 6 ways: $[0, 0, 2]$, $[0, 1, 1]$, $[0, 2, 0]$, $[1, 0, 1]$, $[1, 1, 0]$ and $[2, 0, 0]$.

Input

The only input line has two integers n and m .

Output

Print the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n, m \leq 10^6$

Example

Input:

3 2

Output:

6

CSES Problem Set

Christmas Party

TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

There are n children at a Christmas party, and each of them has brought a gift. The idea is that everybody will get a gift brought by someone else.

In how many ways can the gifts be distributed?

Input

The only input line has an integer n : the number of children.

Output

Print the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^6$

Example

Input:

4

Output:

9

CSES Problem Set

Permutation Order

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Let $p(n, k)$ denote the k th permutation (in lexicographical order) of $1 \dots n$. For example, $p(4, 1) = [1, 2, 3, 4]$ and $p(4, 2) = [1, 2, 4, 3]$.

Your task is to process two types of tests:

1. Given n and k , find $p(n, k)$
2. Given n and $p(n, k)$, find k

Input

The first line has an integer t : the number of tests.

Each test is either "1 n k " or "2 n $p(n, k)$ ".

Output

For each test, print the answer according to the example.

Constraints

- $1 \leq t \leq 1000$
- $1 \leq n \leq 20$
- $1 \leq k \leq n!$

Example

Input:

```
6
1 4 1
1 4 2
2 4 1 2 3 4
2 4 1 2 4 3
1 5 42
2 5 2 4 5 3 1
```

Output:

```
1 2 3 4
1 2 4 3
1
2
2 4 5 3 1
42
```

CSES Problem Set

Permutation Rounds

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a sorted array $[1, 2, \dots, n]$ and a permutation $p_1, p_2, \dots, p_n$. On each round, all elements move according to the permutation: the element at position i moves to position p_i .

After how many rounds is the array sorted again for the first time?

Input

The first line has an integer n .

The next line contains n integers $p_1, p_2, \dots, p_n$.

Output

Print the number of rounds modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input:

8
5 3 2 6 4 1 8 7

Output:

4

Explanation: The array changes as follows after the rounds:

- Round 1: $[6, 3, 2, 5, 1, 4, 8, 7]$
- Round 2: $[4, 2, 3, 1, 6, 5, 7, 8]$
- Round 3: $[5, 3, 2, 6, 4, 1, 8, 7]$
- Round 4: $[1, 2, 3, 4, 5, 6, 7, 8]$

CSES Problem Set

Bracket Sequences I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to calculate the number of valid bracket sequences of length n . For example, when $n = 6$, there are 5 sequences:

- $()()()$
- $()(())$
- $((()))$
- $((())())$
- $((())())$

Input

The only input line has an integer n .

Output

Print the number of sequences modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^6$

Example

Input:

6

Output:

5

CSES Problem Set

Bracket Sequences II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to calculate the number of valid bracket sequences of length n when a *prefix* of the sequence is given.

Input

The first input line has an integer n .

The second line has a string of k characters: the prefix of the sequence.

Output

Print the number of sequences modulo $10^9 + 7$.

Constraints

- $1 \leq k \leq n \leq 10^6$

Example

Input:

6
(()

Output:

2

Explanation: There are two possible sequences: $((()))()$ and $((())())$.

CSES Problem Set**Counting Necklaces**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to count the number of different necklaces that consist of n pearls and each pearl has m possible colors.

Two necklaces are considered to be different if it is not possible to rotate one of them so that they look the same.

Input

The only input line has two numbers n and m : the number of pearls and colors.

Output

Print one integer: the number of different necklaces modulo $10^9 + 7$.

Constraints

- $1 \leq n, m \leq 10^6$

Example

Input:

4 3

Output:

24

CSES Problem Set**Counting Grids**TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

Your task is to count the number of different $n \times n$ grids whose each square is black or white.

Two grids are considered to be different if it is not possible to rotate one of them so that they look the same.

Input

The only input line has an integer n : the size of the grid.

Output

Print one integer: the number of grids modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^9$

Example

Input:

4

Output:

16456

CSES Problem Set**Fibonacci Numbers**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

The Fibonacci numbers can be defined as follows:

- $F_0 = 0$
- $F_1 = 1$
- $F_n = F_{n-2} + F_{n-1}$

Your task is to calculate the value of F_n for a given n .

Input

The only input line has an integer n .

Output

Print the value of F_n modulo $10^9 + 7$.

Constraints

- $0 \leq n \leq 10^{18}$

Example

Input:

10

Output:

55

CSES Problem Set**Throwing Dice**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to calculate the number of ways to get a sum n by throwing dice. Each throw yields an integer between $1 \dots 6$.

For example, if $n = 10$, some possible ways are $3 + 3 + 4$, $1 + 4 + 1 + 4$ and $1 + 1 + 6 + 1 + 1$.

Input

The only input line contains an integer n .

Output

Print the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^{18}$

Example

Input:

8

Output:

125

CSES Problem Set

Graph Paths I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a directed graph that has n nodes and m edges. Your task is to count the number of paths from node 1 to node n with exactly k edges.

Input

The first input line contains three integers n , m and k : the number of nodes and edges, and the length of the path. The nodes are numbered $1, 2, \dots, n$.

Then, there are m lines describing the edges. Each line contains two integers a and b : there is an edge from node a to node b .

Output

Print the number of paths modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 100$
- $1 \leq m \leq n(n - 1)$
- $1 \leq k \leq 10^9$
- $1 \leq a, b \leq n$

Example

Input:

```
3 4 8
1 2
2 3
3 1
3 2
```

Output:

```
2
```

Explanation: The paths are $1 \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 3$ and $1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3$.

CSES Problem Set

Graph Paths II

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a directed weighted graph having n nodes and m edges. Your task is to calculate the minimum path length from node 1 to node n with exactly k edges.

Input

The first input line contains three integers n , m and k : the number of nodes and edges, and the length of the path. The nodes are numbered $1, 2, \dots, n$.

Then, there are m lines describing the edges. Each line contains three integers a , b and c : there is an edge from node a to node b with weight c .

Output

Print the minimum path length. If there are no such paths, print -1 .

Constraints

- $1 \leq n \leq 100$
- $1 \leq m \leq n(n - 1)$
- $1 \leq k \leq 10^9$
- $1 \leq a, b \leq n$
- $1 \leq c \leq 10^9$

Example

Input:

```
3 4 8
1 2 5
2 3 4
3 1 1
3 2 2
```

Output:

```
27
```

CSES Problem Set

Sum of Four Squares

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A well known result in number theory is that every non-negative integer can be represented as the sum of four squares of non-negative integers.

You are given a non-negative integer n . Your task is to find four non-negative integers a, b, c and d such that $n = a^2 + b^2 + c^2 + d^2$.

Input

The first line has an integer t : the number of test cases.

Each of the next t lines has an integer n .

Output

For each test case, print four non-negative integers a, b, c and d that satisfy $n = a^2 + b^2 + c^2 + d^2$.

Constraints

- $1 \leq t \leq 1000$
- $0 \leq n \leq 10^7$
- the sum of all n is at most 10^7

Example

Input:

```
3
5
30
322266
```

Output:

```
2 1 0 0
1 2 3 4
314 159 265 358
```

CSES Problem Set**Triangle Number Sums**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A triangle number is a positive integer of the form $1 + 2 + \dots + k$. The first triangle numbers are 1, 3, 6, 10 and 15.

Every positive integer can be represented as a sum of triangle numbers. For example, $42 = 21 + 21$ and $1337 = 1326 + 10 + 1$.

Given a positive integer n , determine the smallest number of triangle numbers that sum to n .

Input

The first line has an integer t : the number of tests.

After that, each line has a positive integer n .

Output

For each test, print the smallest number of triangle numbers.

Constraints

- $1 \leq t \leq 100$
- $1 \leq n \leq 10^{12}$

Example

Input:

```
5
1
2
3
42
1337
```

Output:

```
1
2
1
2
3
```

CSES Problem Set

Dice Probability

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You throw a dice n times, and every throw produces an outcome between 1 and 6. What is the probability that the sum of outcomes is between a and b ?

Input

The only input line contains three integers n , a and b .

Output

Print the probability rounded to six decimal places (rounding half to even).

Constraints

- $1 \leq n \leq 100$
- $1 \leq a \leq b \leq 6n$

Example

Input:

2 9 10

Output:

0.194444

CSES Problem Set

Moving Robots

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Each square of an 8×8 chessboard has a robot. Each robot independently moves k steps, and there can be many robots on the same square.

On each turn, a robot moves one step left, right, up or down, but not outside the board. It randomly chooses a direction among those where it can move.

Your task is to calculate the expected number of *empty* squares after k turns.

Input

The only input line has an integer k .

Output

Print the expected number of empty squares rounded to six decimal places (rounding half to even).

Constraints

- $1 \leq k \leq 100$

Example

Input:

10

Output:

23.120740

CSES Problem Set**Candy Lottery**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n children, and each of them independently gets a random integer number of candies between 1 and k .

What is the expected maximum number of candies a child gets?

Input

The only input line contains two integers n and k .

Output

Print the expected number rounded to six decimal places (rounding half to even).

Constraints

- $1 \leq n \leq 100$
- $1 \leq k \leq 100$

Example

Input:

2 3

Output:

2.444444

CSES Problem Set**Inversion Probability**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

An array has n integers $x_1, x_2, \dots, x_n$, and each of them has been randomly chosen between 1 and r_i . An inversion is a pair (a, b) where $a < b$ and $x_a > x_b$.

What is the expected number of inversions in the array?

Input

The first input line contains an integer n : the size of the array.

The second line contains n integers $r_1, r_2, \dots, r_n$: the range of possible values for each array position.

Output

Print the expected number of inversions rounded to six decimal places (rounding half to even).

Constraints

- $1 \leq n \leq 100$
- $1 \leq r_i \leq 100$

Example

Input:

3
5 2 7

Output:

1.057143

CSES Problem Set**Stick Game**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a game where two players remove sticks from a heap. The players move alternately, and the player who removes the last stick wins the game.

A set $P = \{p_1, p_2, \dots, p_k\}$ determines the allowed moves. For example, if $P = \{1, 3, 4\}$, a player may remove 1, 3 or 4 sticks.

Your task is find out for each number of sticks $1, 2, \dots, n$ if the first player has a winning or losing position.

Input

The first input line has two integers n and k : the number of sticks and moves.

The next line has k integers $p_1, p_2, \dots, p_k$ that describe the allowed moves. All integers are distinct, and one of them is 1.

Output

Print a string containing n characters: W means a winning position, and L means a losing position.

Constraints

- $1 \leq n \leq 10^6$
- $1 \leq k \leq 100$
- $1 \leq p_i \leq n$

Example

Input:

```
10 3
1 3 4
```

Output:

```
WLWWWLWLW
```

CSES Problem Set**Nim Game I**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n heaps of sticks and two players who move alternately. On each move, a player chooses a non-empty heap and removes any number of sticks. The player who removes the last stick wins the game.

Your task is to find out who wins if both players play optimally.

Input

The first input line contains an integer t : the number of tests. After this, t test cases are described:

The first line contains an integer n : the number of heaps.

The next line has n integers $x_1, x_2, \dots, x_n$: the number of sticks in each heap.

Output

For each test case, print "first" if the first player wins the game and "second" if the second player wins the game.

Constraints

- $1 \leq t \leq 2 \cdot 10^5$
- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- the sum of all n is at most $2 \cdot 10^5$

Example

Input:

```
3
4
5 7 2 5
2
4 1
3
3 5 6
```

Output:

```
first
first
second
```

CSES Problem Set**Nim Game II**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n heaps of sticks and two players who move alternately. On each move, a player chooses a non-empty heap and removes 1, 2, or 3 sticks. The player who removes the last stick wins the game.

Your task is to find out who wins if both players play optimally.

Input

The first input line contains an integer t : the number of tests. After this, t test cases are described:

The first line contains an integer n : the number of heaps.

The next line has n integers $x_1, x_2, \dots, x_n$: the number of sticks in each heap.

Output

For each test case, print "first" if the first player wins the game and "second" if the second player wins the game.

Constraints

- $1 \leq t \leq 2 \cdot 10^5$
- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- the sum of all n is at most $2 \cdot 10^5$

Example

Input:

```
3
4
5 7 2 5
2
4 1
3
4 4 4
```

Output:

```
first
first
second
```

CSES Problem Set

Stair Game

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a staircase consisting of n stairs, numbered $1, 2, \dots, n$. Initially, each stair has some number of balls.

There are two players who move alternately. On each move, a player chooses a stair k where $k \neq 1$ and it has at least one ball. Then, the player moves any number of balls from stair k to stair $k - 1$. The player who moves last wins the game.

Your task is to find out who wins the game when both players play optimally.

Note that if there are no possible moves at all, the second player wins.

Input

The first input line has an integer t : the number of tests. After this, t test cases are described:

The first line contains an integer n : the number of stairs.

The next line has n integers $p_1, p_2, \dots, p_n$: the initial number of balls on each stair.

Output

For each test, print "first" if the first player wins the game and "second" if the second player wins the game.

Constraints

- $1 \leq t \leq 2 \cdot 10^5$
- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq p_i \leq 10^9$
- the sum of all n is at most $2 \cdot 10^5$

Example

Input:

```
3
3
0 2 1
4
1 1 1 1
2
5 3
```

Output:

```
first
second
first
```

CSES Problem Set

Grundy's Game

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a heap of n coins and two players who move alternately. On each move, a player chooses a heap and divides into two nonempty heaps that have a different number of coins. The player who makes the last move wins the game.

Your task is to find out who wins if both players play optimally.

Input

The first input line contains an integer t : the number of tests.

After this, there are t lines that describe the tests. Each line has an integer n : the number of coins in the initial heap.

Output

For each test case, print "first" if the first player wins the game and "second" if the second player wins the game.

Constraints

- $1 \leq t \leq 10^5$
- $1 \leq n \leq 10^6$

Example

Input:

```
3
6
7
8
```

Output:

```
first
second
first
```

CSES Problem Set

Another Game

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n heaps of coins and two players who move alternately. On each move, a player selects some of the nonempty heaps and removes one coin from each heap. The player who removes the last coin wins the game.

Your task is to find out who wins if both players play optimally.

Input

The first input line contains an integer t : the number of tests. After this, t test cases are described:

The first line contains an integer n : the number of heaps.

The next line has n integers $x_1, x_2, \dots, x_n$: the number of coins in each heap.

Output

For each test case, print "first" if the first player wins the game and "second" if the second player wins the game.

Constraints

- $1 \leq t \leq 2 \cdot 10^5$
- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- the sum of all n is at most $2 \cdot 10^5$

Example

Input:

```
3
3
1 2 3
2
2 2
4
5 5 4 5
```

Output:

```
first
second
first
```